Code

Stats 20, F23 – Homework 6

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Question 1

(a)

```
reading_ease <- function(x) {
  if (length(x) > 1) {
    sentences <- paste(unlist(x), collapse = " ")
  } else {
    sentences <- paste(x, collapse = " ")
  }
  sentences <- strsplit(sentences, split = "[.!?:;]")[[1]]
  sentences <- tolower(sentences)
  sentences <- gsub(pattern = "[[:punct:]]", replacement = "",
sentences)

  words <- strsplit(sentences, split = " ")

  keep <- function(w) {
    w[nchar(w) > 0]
  }

  words <- lapply(words, keep)
  ASL = length(unlist(words))/length(unlist(sentences))

  is_special_ending <- function(ending) {
    is_es <- all(ending == c("e","s"))
    is_ed <- all(ending == c("e","d"))
    is_e_not_le <- ending[2] == "e" & ending[1] != "l"
    is_es | is_ed | is_e_not_le
  }

  rm_special_endings <- function(word_letters) {
    word_tail <- tail(word_letters, n = 2)
    if (is_special_ending(word_tail)) {
      if (word_tail[2] == "e") {
        word_letters[-length(word_letters)]
      } else {
        head(word_letters, n = -2)
      }
    } else {
      word_letters
    }
  }

  is_vowel <- function(letter) {
    letter %in% c("a", "e", "i", "o", "u", "y")
  }

  count_syl <- function(word) {
    word_letters <- unlist(strsplit(word,split = ""))
    if (length(word_letters) <= 3) {
      1
    } else {
      word_letters <- rm_special_endings(word_letters)
      word_vowels <- is_vowel(word_letters)
```

```
      sum(word_vowels) - sum(diff(which(word_vowels)) == 1)
    }
  }

  syllables = sum(unlist(lapply(unlist(words),count_syl)))

  ASW = syllables/length(unlist(words))

  206.835 - (1.015 * ASL) - (84.6 * ASW)
}
```

(b)

```
waffles_vec <- c("We need to remember what's important in life:
friends, waffles, work.",
                 "Or waffles, friends, work.",
                 "Doesn't matter, but work is third.")
reading_ease(waffles_vec)
```

```
## [1] 96.76339
```

Question 2

```
my_nchar <- function(x) {
  chars = c()
  index = 1
  for (i in x) {
    chars[index] <- length(unlist(strsplit(i, split = "")))
    index <- index + 1
  }
  chars
}
```

Question 3

```
my_strrep <- function(x,times)  {
  output <- rep(x,times)
  paste(output,sep = "",collapse="")
}
```

Question 4

```
`%is_in%` <- function(x, table) {
  index <- 1
  result <- c()
  for (i in x) {
    result[index] <- (any(table == i))
    index <- index + 1
  }
  result
}
```

Question 5

```
my_gregexpr <- function(pattern,text) {
  length <- nchar(pattern)
  chars <- unlist(strsplit(text,split = ""))
  output <- as.list(rep(-1, length(chars)))
  for (i in seq_along(chars)) {
    substring <- paste(chars[i:(i + length - 1)],collapse = "")
    if (substring == pattern) {
      output[[i]] <- i
    }

  }
  output
}
```